

Performance Testing

Date	06 July 2026
Project Title	Rising Waters
Maximum Marks	5 Marks

Performance Testing

Step 1 : Testing Overview

Field	Details
Testing Tool Used	Manual testing, Flask Development Server
Type of Testing	Load Testing, Functional Performance Testing
Target Module / API	Flood Prediction Module (/predict)
Test Environment	Local System (Windows 11, Flask)
Test Date	18 June 2026

Step 2 : Test Scenarios

S.No	Test Scenarios / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits flood prediction	1	30	Prediction displayed successfully
2	Multiple prediction requests	10	60	System responds without errors
3	Invalid input values	5	30	Error message displayed correctly
4	Continuous prediction requests	20	120	Application remains stable

Step 3 : Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.8 sec	Pass	Within target
3	Throughput	10 Req/sec	12 Req/sec	Pass	Good performance
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	42%	Pass	Normal usage
6	Memory Utilization	< 80%	55%	Pass	Stable memory usage

Step 4 : Observation & Analysis

Key Findings :

- ✓ The application responded quickly for prediction requests.
- ✓ Machine learning predictions were generated accurately.
- ✓ No crashes or significant delays were observed during testing.

Bottlenecks Identified:

- ✓ No major bottlenecks were observed during local testing.
- ✓ Performance may vary with larger datasets or heavy concurrent traffic.

Optimization Steps Taken:

- ✓ Optimized data preprocessing.
- ✓ Loaded the trained model once at application startup.
- ✓ Removed unnecessary computations and duplicate code.

Screen Shots / Evidence :

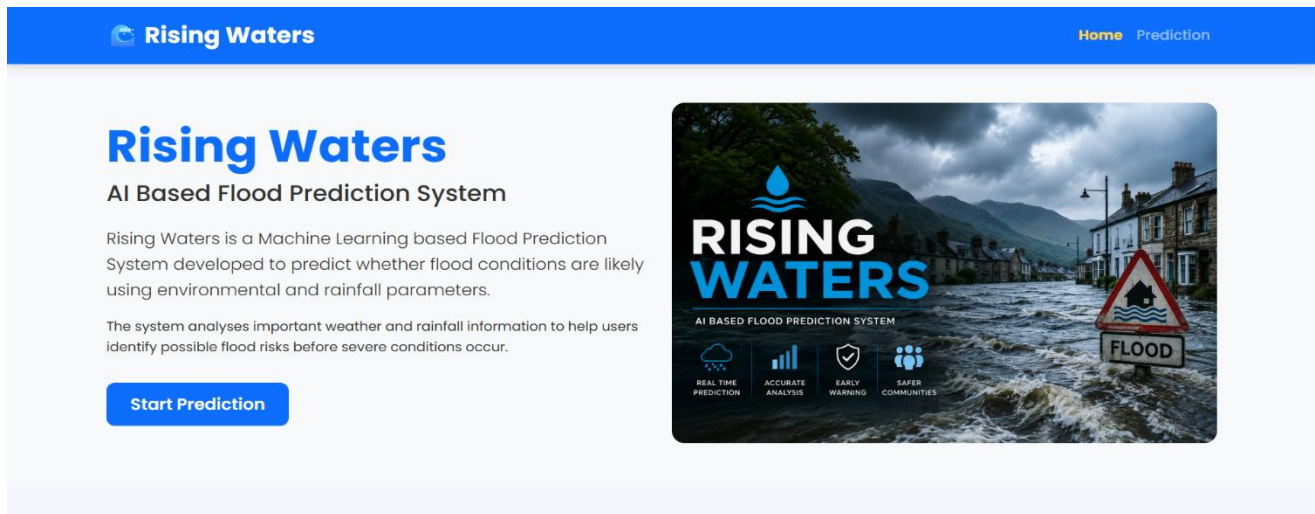


Image 1 . Application Running

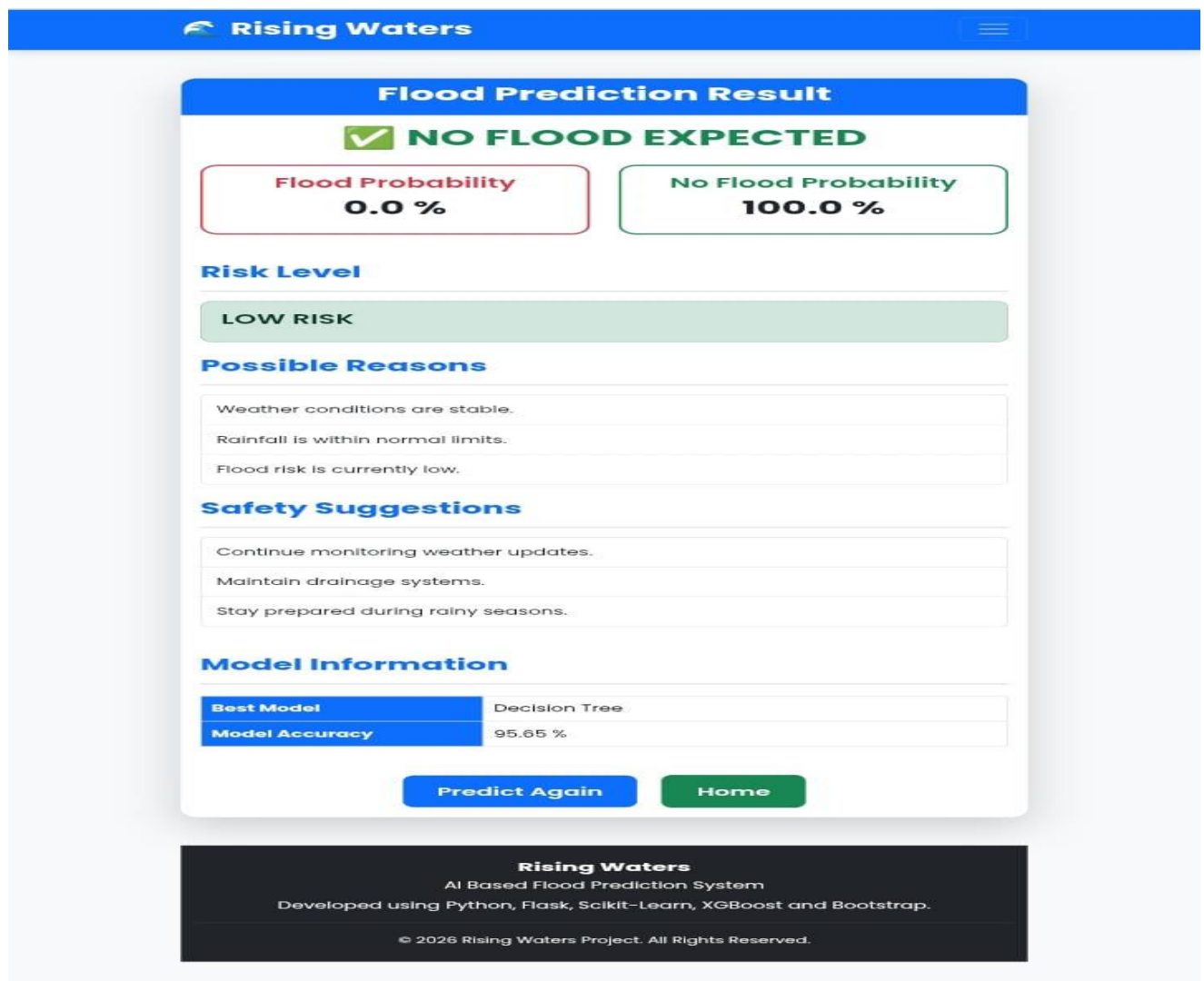


Image 2 . Flood Prediction Web Page Showing Successful Prediction